Yiyang Chen

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Education

University of California, Riverside **Riverside, CA**

Exchange Student – Graduate Preparation Program *Sept. 2025 – June 2026*

- **CS Coursework:** Intermediate Data Structures & Algorithms (A+), ML & Data Mining (A+), Artificial Intelligence (A+), Deep Learning (A), Discrete Mathematics (A+), C++ for Programmers (A+), Problem Solving Lab (A)
- **Research & Engineering:** Research for Undergraduates (A+), Structural Ordering in Colloidal Dispersions (A+), Chemical Process Analysis (A)

East China University of Science and Technology (ECUST) **Shanghai, China**

B.Eng. in Chemical Engineering – GPA: 3.68/4.00 *Sept. 2022 – June 2026*

- **CS & ML Coursework:** Data Structures & Algorithms (94), Big Data Algorithms (100), Machine Learning (98), Fundamentals of Programming (98), Computer Networks & Cloud Computing (92), AI Technology & Application (91), Operating Systems (85), Big Data Visualization (91)
- **Applied Computing:** Computer Applications to Chemical Engineering (82), Process Control and Instruments (89)
- **Mathematics:** Advanced Calculus I (98), Probability & Mathematical Statistics (90)

University of Texas at Austin **Austin, TX**

Summer Visiting Student *July 2024 – Aug. 2024*

- Data Analysis (R), Python Programming, Academic English; all A grades

Research Experience

Functional-Group Graph Networks for Solubility Prediction **UC Riverside**

Undergraduate Researcher – Advisor: Prof. Jianzhong Wu *Jan. 2026 – Present*

- Implemented an exact-cover search that resolves overlapping SMARTS matches into non-overlapping functional groups. Across 10,680 molecules, this reduced graph size by 23.4% on average.
- Built separate four-layer GIN encoders for solutes and solvents. A three-coefficient readout uses their representations to predict solubility at different temperatures.
- Evaluated on 182,608 measurements using pair-disjoint splits, keeping repeated temperatures for the same solute–solvent pair out of both training and test sets ($R^2 = 0.864 \pm 0.014$; MAE 0.313 ± 0.013).
- Packaged the trained models, SMILES preprocessing, and command-line prediction into a pip-installable repository, so others can run the model without rebuilding the training pipeline.

Selected Computer Science Projects

TC-SS-DPMixer: Energy Forecasting | *UCR CS228* | Code **2026**

- A zero-parameter persistence forecast beat several neural models in the first benchmark. Traced the discrepancy to training with MSE while evaluating with MAE, then reran the comparisons with matched losses.
- Built TC-SS-DPMixer with hour-dependent branch weights and a separate path for appliance-use spikes. It reached 34.83 Wh MAE with 68K parameters; iTransformer reached 34.19 Wh with 412K.
- Removed individual branches, varied the parameter budget, and plotted the learned gates by hour. Their peaks tracked the hours when the persistence forecast made its largest errors.
- Pretrained PatchTST on masked time-series patches. With 20% of labels, MAE dropped from 63.9 to 54.1 Wh; with the full labeled set, the same pretraining did not help.

VoidTalk – Anonymous Discussion Platform | *RoseHack 2026* | Code | Devpost **2026**

- Built the posting and reply screens in a three-person RoseHack team, connecting the Next.js interface to Supabase for anonymous posts and private conversations.
- Proposed ranking replies by what the author wanted—advice, empathy, disagreement, or a journal entry. Wrote the mode-specific prompts and a Flask endpoint that sorts replies into three visibility tiers.

Feature Selection and Heuristic Search | *UCR CS170* | Code **2026**

- Implemented forward selection and backward elimination using 1-NN leave-one-out validation. Two of 64 features reached 97.3% accuracy on 3,000 samples, versus 68.0% with all 64.
- Wrote uniform-cost search and A* for the 8-puzzle, comparing Manhattan-distance and misplaced-tile heuristics by expanded nodes and queue size.

Manuscripts

- **Yiyang Chen**, Jianzhong Wu. "Thermodynamics-Guided Functional Group Graph Neural Network for Molecular Solubility Prediction." Submitted to *Journal of Chemical Information and Modeling*, 2026.
- **Yiyang Chen** et al. "Coking-aware multi-objective scheduling of ethylene cracking furnace clusters." Resubmitted to *Industrial & Engineering Chemistry Research*, 2026.

Hackathon & Software Projects

FluxMind – Student Workload Planner | *Citrus Hack 2026* | Devpost **2026**

- Built a planner that turns a student's deadlines, available hours, and competing commitments into three alternative weekly plans, each showing what must be kept, delayed, or dropped.
- Wired the Next.js interface to a structured analysis API, mapping returned risk scores and plan breakdowns into the comparison screen.
- Kept the demo usable when Gemini or ElevenLabs calls failed by falling back to local example results instead of leaving users with an empty screen.

TerraPulse – 3D Travel Explorer | *DesignVerse 2026* | Devpost **2026**

- Built a Three.js globe with 78 selectable cities. Selecting a marker opens a city page with places to visit, food, weather, and community posts.
- Designed the city cards, landmark views, and climate-dependent color schemes, turning the globe into a travel site rather than a standalone visualization.
- Added an image pipeline that caches destination photos and generates climate-specific placeholders when a city or landmark image is missing.

EcoLung – Air-Quality Exposure Simulator | *BearHacks* | Code **2026**

- Connected OpenAQ air-quality data to a FastAPI service and a React/Three.js interface that shows how exposure changes with location and daily routine.
- Added side-by-side comparisons for changes such as time outdoors or mask use; the 3D lung updates with the selected scenario.
- Added fallback sample data so the visualization could still run if the external air-quality feed was unavailable.

Work & Internship Experience

University of California, Riverside – Professional and Global Education **Riverside, CA**

Student Assistant, Global Professional Education *Dec. 2025 – June 2026*

- Prepared 26 UCR program documents for a NotebookLM adviser embedded in GEN 884, bringing scattered enrollment and program rules into one searchable source set.
- Tested the adviser with questions international students actually ask. When an answer was missing or misleading, revised the supporting documents and prompts, then tested it again.
- Wrote short guidance-video scripts and checked international students' course requests for conflicts and missing enrollment information.

University of California, Riverside – Professional and Global Education **Riverside, CA**

Intern, Global Professional Education *Sept. 2026 – Present*

- Reworked advising prompts and source materials to make answers about course selection and program procedures easier to verify.
- Prepared student-facing explanations for use alongside the online advising assistant, working with the instructional design team on confusing or incomplete responses.

Research Interests

Graph representation learning, efficient neural networks, self-supervised learning, and machine learning for scientific data.

Technical Skills

Programming: Python, C++, R, TypeScript, JavaScript

ML Libraries: PyTorch, PyTorch Geometric, scikit-learn, RDKit, NumPy, pandas

Algorithms & Data Structures: Divide-and-conquer, greedy methods, dynamic programming, graph traversal (BFS/DFS), Dijkstra's algorithm, trees, heaps, hashing, A* and exact-cover search

ML Methods: Graph neural networks (GIN), Transformers, self-supervised pretraining, time-series forecasting, feature selection, cross-validation, ablation studies

Development: Git, React, Next.js, Supabase, PostgreSQL, FastAPI, Flask, Three.js, Vercel

Languages: Chinese (Native), English (CET-4: 558, CET-6: 480)

Honors & Awards

Pudding Success On Award (Innovative Product Ideas), UCR DesignVerse 2026
First Prize, National Energy Economics Competition, China 2025
Academic Scholarship, Third Class 2022–2023, 2024–2025
Mathematical Modeling Competition, Provincial Third Prize 2024
Mathematics Competition, Provincial Third Prize 2024
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